

Machine Learning - Full Course Notes

Module 1: Introduction

What is ML, AI vs ML vs DL, applications, workflow, supervised/unsupervised/reinforcement learning.

Module 2: Python

NumPy, Pandas, Matplotlib, data preprocessing, feature engineering.

Module 3: Statistics

Mean, median, variance, probability, distributions, hypothesis testing.

Module 4: Regression

Linear, Multiple, Polynomial, Ridge, Lasso, ElasticNet.

Module 5: Classification

Logistic Regression, KNN, Naive Bayes, SVM, Decision Trees.

Module 6: Ensemble

Random Forest, Bagging, Boosting, XGBoost overview.

Module 7: Clustering

K-Means, Hierarchical, DBSCAN.

Module 8: Dimensionality Reduction

PCA, t-SNE overview.

Module 9: Model Evaluation

Train/Test split, Cross Validation, Accuracy, Precision, Recall, F1, ROC-AUC.

Module 10: Deep Learning

Neural Networks, TensorFlow, Keras basics.

Module 11: NLP

Text preprocessing, TF-IDF, embeddings, sentiment analysis.

Module 12: Computer Vision

OpenCV basics, CNN introduction.

Module 13: Deployment

Flask/FastAPI, Docker basics, cloud deployment.

Projects

House Price Prediction, Spam Detection, Customer Churn, Image Classification, Recommendation System.